

Question:

What are the minimum and maximum possible values of Glasgow Coma Score?

Answer Choices:

- A. Minimum = 3, Maximum = 15
- B. Minimum = 0, Maximum = 13
- C. Minimum = 0, Maximum = 15
- D. Minimum = 3, Maximum = 18

Reasoning + Answer:

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Finding Reasoning Paths :

- 1 . **Definition of Glasgow Coma Scale (GCS)**: The GCS assesses consciousness based on three components : eye response , verbal response , and motor response , assigning scores from 3 to 15 .
- 2 . **Understanding the Range**: Explore why the lowest possible score (3) and the highest possible score (15) are achieved and what they represent in terms of consciousness level .
- 3 . **Clinical Context**: Consider the clinical scenario where a patient with locked-in syndrome might have a GCS of 15 (e.g ., Shirley 's syndrome), but may not be responsive to stimuli like blinking .

Reasoning Process :

- 1 . **Understanding the GCS**: The GCS is a scale used to assess the level of consciousness in patients . It is based on three components : eye response (E), verbal response (V), and motor response (M). Each component is scored from 1 to 5 , and the total score ranges from 3 to 15 . A score of 3 indicates a coma or deepening coma , while a score of 15 indicates full consciousness , including eye-opening to painful stimuli , verbal response , and purposeful movements .
- 2 . **Minimum Score (3)**: A score of 3 indicates a coma or very deep coma . This could occur in scenarios such as stroke , traumatic brain injury , or advanced diseases like ALS or amyotrophic lateral sclerosis . In such cases , the patient may have no spontaneous eye response (E = 1), no verbal response (V = 1), and no motor response (M = 1).
- 3 . **Maximum Score (15)**: A score of 15 indicates full consciousness , including responsiveness to painful stimuli (E ≥ 4), verbal response (V ≥ 4), and purposeful movements (M ≥ 3). This is consistent with a patient who appears awake and alert , albeit possibly with a locked-in state (e.g ., Shirley 's syndrome), where the patient cannot use motor responses due to severe paralysis .
- 4 . **Clinical Context**: Shirley 's syndrome describes a condition where a patient with locked-in syndrome can open their eyes to painful stimuli (E ≥ 4) but cannot use the other components of the GCS , resulting in a total score of 15 .

Conclusion :

The minimum possible Glasgow Coma Score is **3** , representing a coma or deepening coma , and the maximum possible score is **15** , representing full consciousness , including responsiveness to painful stimuli .

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<answer>

Minimum = 3 , Maximum = 15

</answer>